LAB EXPERIMENT

COURSE CODE: ITA0318

COURSE NAME: MOBILE COMPUTING

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EXP -01

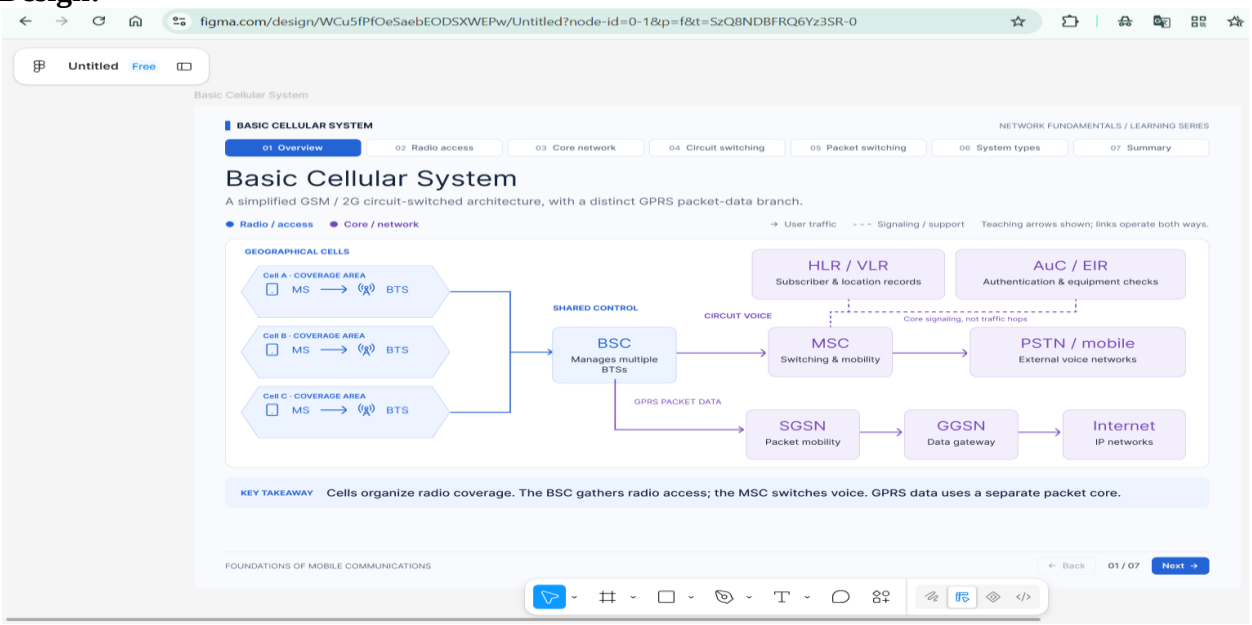
1.Use Figma to create an interactive diagram illustrating the components of a basic cellular system.

Aim: To create an interactive diagram illustrating the components of a basic cellular system

Procedure: -

1. openFigma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
11. Review and edit the Prototype
12. Save and Share

Design: -



From the handset to the radio network

BASIC CELLULAR SYSTEM

NETWORK FUNDAMENTALS / LEARNING SERIES

01 Overview | 02 Radio access | 03 Core network | 04 Circuit switching | 05 Packet switching | 06 System types | 07 Summary

From the handset to the radio network

Radio access connects moving users to fixed infrastructure, one local coverage area at a time.

CELL · A GEOGRAPHICAL AREA, NOT A HOP

MS
Mobile station

Radio link

BTS
Radio transmitter / receiver

Backhaul

BSC
Controls a group of BTSs

MOVING BETWEEN CELLS

As a user moves, a handover can transfer the radio connection to another cell.

MS

Mobile Station

The user device and subscriber identity (SIM) that access the mobile network.

Example: a mobile phone making a call.

Cells

Geographical coverage

Local radio coverage areas served by base stations. Real cell shapes vary with terrain and radio conditions.

Cells allow frequencies to be reused across a network.

BTS

Base Transceiver Station

Transmits and receives radio signals, linking mobile stations to the radio access network.

One physical site can serve several sectors or cells.

BSC

Base Station Controller

Manages several BTSs, allocates radio resources and coordinates many handovers.

Connects the radio domain toward the core network.

KEY TAKEAWAY A cell is a place; a BTS is equipment. The BSC coordinates the radio resources used by many BTSs.

The core switches calls and knows the subscriber

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The core switches calls and knows the subscriber

Traffic passes through switching nodes. Subscriber, security and equipment functions support decisions through signaling.

MSC

Mobile Switching Centre

Sets up and switches circuit calls, routes voice traffic and coordinates mobility in the core.

ON THE VOICE PATH

BSC → MSC → PSTN or another mobile network

HLR

Home Location Register

Stores the subscriber's service profile and a reference to their current serving location.

VLR

Visitor Location Register

Holds temporary subscriber and location information for users in the current serving region.

AuC

Authentication Centre

Supplies authentication and key material to help verify the subscriber and secure the radio link.

EIR

Equipment Identity Register

Checks the device's equipment identity (IMEI) and status, such as whether it is barred.

EXTERNAL NETWORKS

PSTN / Internet

PSTN: the public switched telephone network. Reached from the MSC for circuit voice calls.

Internet: external IP networks. GPRS reaches them through SGSN / GGSN, not through the MSC.

KEY TAKEAWAY The MSC carries circuit calls; HLR, VLR, AuC and EIR provide information and checks, not serial traffic hops.

Circuit switching: reserve a path for the call

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Circuit switching: reserve a path for the call

A logical circuit is established before conversation and remains allocated until the call ends.

GSM CIRCUIT VOICE · USER TRAFFIC

Conversation is bidirectional

MS — User device → BTS — Radio connection → BSC — Radio control → MSC — Call switching → PSTN / mobile — Destination network

Capacity stays assigned during silence

Call established | Time → | Call released

Circuit A: One call · reserved end to end

Resources can be time slots or channels—not necessarily a separate physical wire. Other users cannot use this reserved capacity for the duration of the call.

1. **Set up**
   Signaling locates the user and allocates a circuit.
2. **Carry the conversation**
   Voice uses the reserved path with predictable capacity.
3. **Release**
   The call ends and resources become available again.

KEY TAKEAWAY Circuit switching reserves capacity for one call—even when neither person is speaking.

FOUNDATIONS OF MOBILE COMMUNICATIONS

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Packet switching: share links, route packets

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Packet switching: share links, route packets

GPRS adds a packet domain to GSM radio access. Internet user traffic does not pass through the MSC.

GSM / GPRS · PACKET-DATA BRANCH

Simplified: packet-control functions not shown

MS — Packet-capable device → BTS / BSC — Shared radio access → SGSN — Serving packet node → GGSN — Gateway to IP networks → Internet — External IP network

Many users, one shared link

A1 | B1 | A2 | B2 | A3 | B3 →

● User A ● User B

Data is split into packets and forwarded across shared links. Capacity is used as needed; there is no dedicated end-to-end circuit.

SGSN

Serving GPRS Support Node

Tracks packet-user mobility and sessions, and forwards data between radio access and the GGSN.

GGSN

Gateway GPRS Support Node

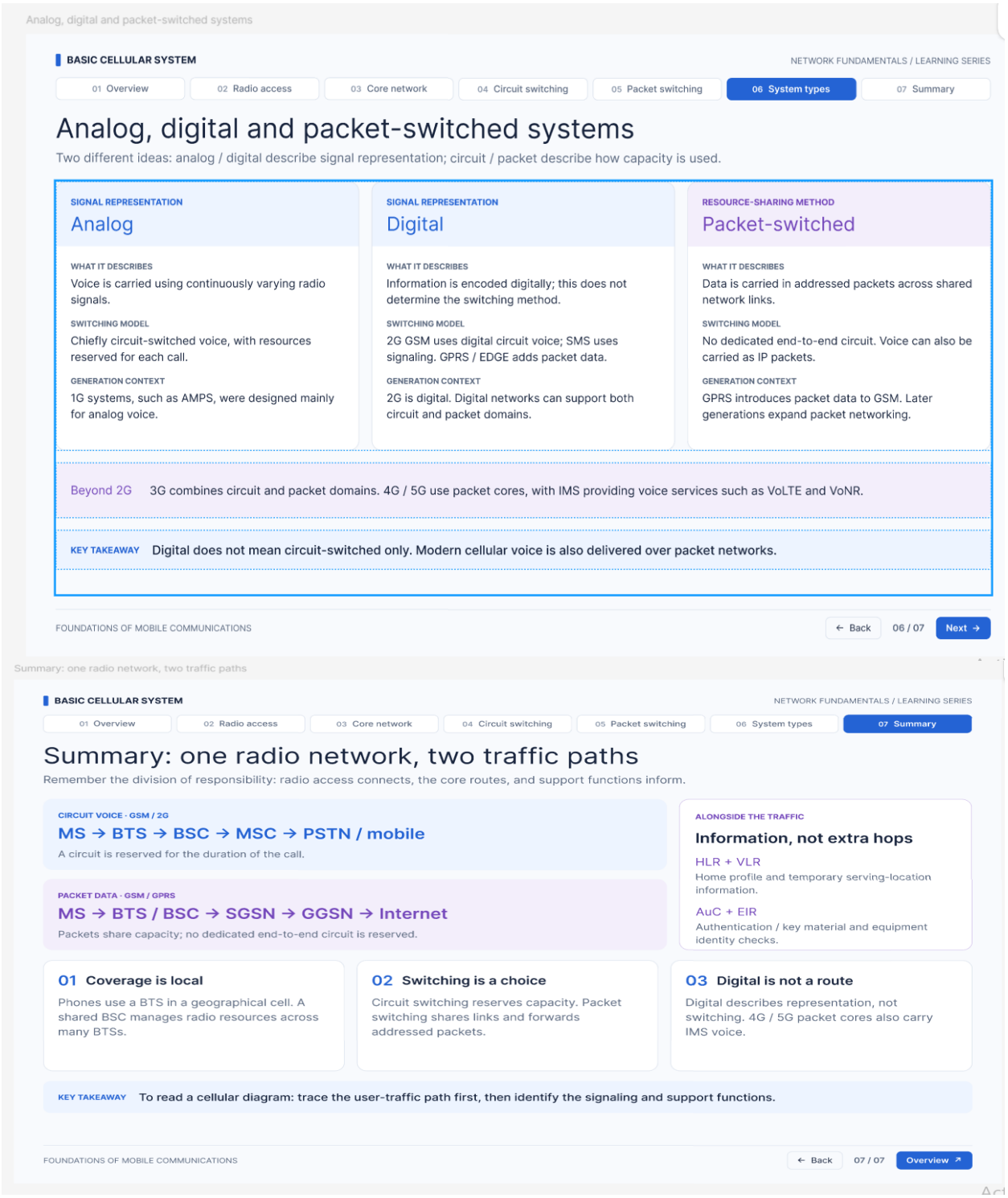
Connects the GPRS packet core to external IP networks and routes packets to and from the Internet.

KEY TAKEAWAY Packet switching shares capacity across users. In GPRS, the SGSN and GGSN—not the MSC—form the Internet path.

FOUNDATIONS OF MOBILE COMMUNICATIONS

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Summary: one radio network, two traffic paths



Result: -

Hence an interactive diagram illustrating the components of a basic cellular system using Figma is created and executed successfully